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Efficiency Analysis Of an Algorithm Using Asymptotic Notations

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Problem Definition

- A computer system processes a large amount of data using a specific algorithm.
- The size of the input is represented by n , which affects execution time.
- As the input size increases, the algorithm requires more processing time.
- The main objective is to evaluate the algorithm's performance for large datasets.
- Asymptotic notations help predict the efficiency without actual execution.



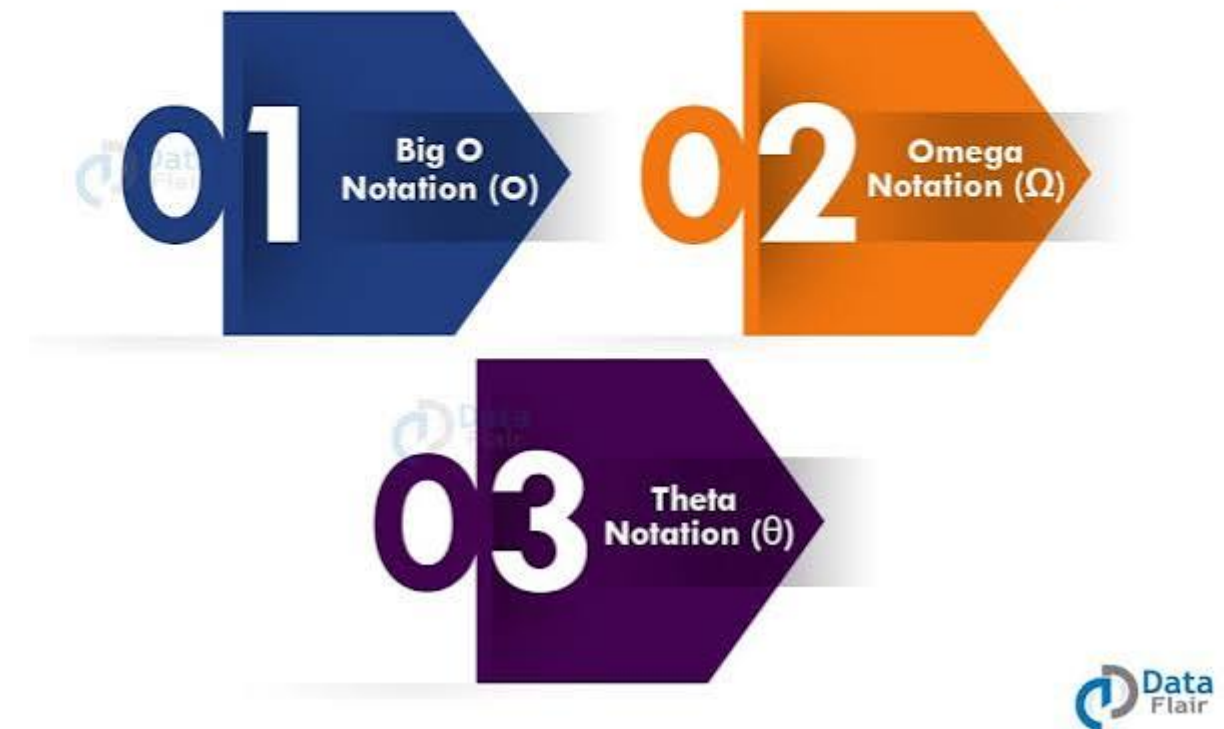


Asymptotic Notations



- Big O (O) describes the maximum execution time in the worst-case scenario.
- Big Omega (Ω) represents the minimum execution time in the best case.
- Big Theta (Θ) gives the average or exact growth rate of the algorithm.
- These notations help compare different algorithms for the same problem.
- They are widely used to measure time complexity and scalability.

Asymptotic Notations





Performance Analysis

- Consider Linear Search , where each element is checked one by one.
- The best – case complexity is $O(1)$ when the element is found immediately.
- The average-case complexity is $O(n)$ because several elements may be checked.
- The worst-case complexity is $O(n)$ when the element is at the end or absent.
- This analysis helps estimate performance for different input sizes.



Execution Trend

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- The execution time increases as the input size becomes larger.
 - Algorithms with lower time complexity perform better on large datasets.
 - $O(n \log n)$ algorithms are generally more efficient than $O(n^2)$ algorithms.
 - A graph of Input Size (n) versus Execution Time clearly shows this trend.
 - Efficient algorithms improve scalability and reduce processing time.



CONCLUSION



- Asymptotic notations provide a simple way to analyze algorithm efficiency.
- They help predict performance without implementing the algorithm.
- Mathematical analysis is useful for comparing different algorithms.
- Efficient algorithms can process large datasets with better scalability.
- Choosing the right algorithm improves system performance and saves computational resources.